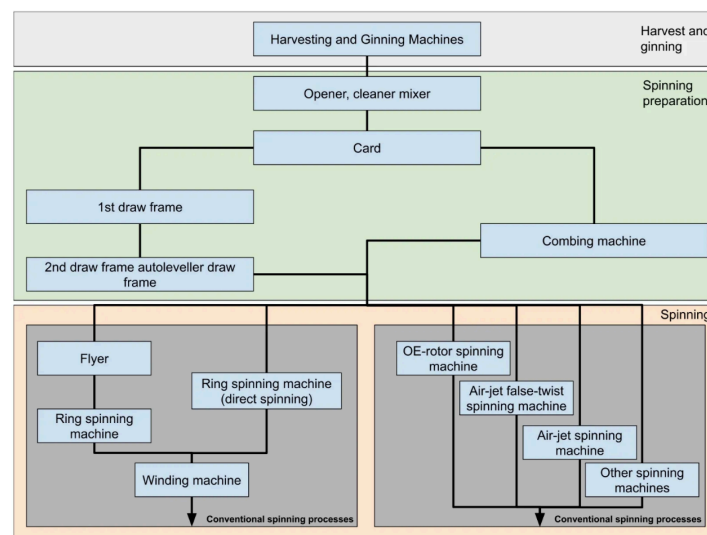




Spinning - Ring Spun

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1. Introduction



1.1 Purpose of the study

This document describes the life cycle inventory (LCI) model for ring-spun yarn production as part of the Carbonfact Textile LCA Database. The aim is to provide a transparent, literature-based, and reproducible foreground inventory covering 15 fibre families across a continuous range of yarn fineness (45–600 dtex), yielding 875 individual process datasets. The model is designed to replace existing generic spinning dataset proxies with parameterised LCIs that are scientifically robust, independently verifiable, and suitable for open-source publication on GitHub.

The work involves a structured literature review, regression modelling, blended curve construction, and external validation, followed by documentation of all non-electricity parameters (lubrication, process water, yarn losses, capital goods, plastic bobbins) and a full LCIA comparison against EF 3.1 reference datasets.

1.2 Spinning process - Ring spun

Ring spinning is the most widely deployed yarn manufacturing technology globally, producing the majority of staple-fibre yarns from cotton, wool, synthetic staple, and blended feedstocks. The process converts loose fibres (in the form of a roving or sliver) into a twisted continuous yarn by simultaneously drafting the material to the target fineness and inserting twist via a traveller rotating on a ring. The resulting ring-spun yarn is characterised by protruding fibre ends, a soft hand, and a relatively hairy surface, properties that distinguish it from open-end rotor-spun and air-jet-spun alternatives.

The sub-processes included within the system boundary are described in the table below.

Sub-process	Function	Primary energy driver
Opening and cleaning	Loosens compressed bales; removes trash, short fibres, and seed coat fragments	Beater and fan motors; largely independent of yarn fineness
Carding	Aligns individual fibres into a thin web (sliver); removes remaining short fibres and neps	Card cylinder and doffer drive; dominant preparation stage for cotton
Combing (combed routes only)	Combs out fibres shorter than a defined staple length (noil removal); produces a more regular, lustrous sliver	Combing cylinder drive; responsible for higher waste fraction in combed cotton routes (~12–15% of input)
Drawing (breaker and finisher)	Drafts and doubles slivers to improve evenness and fibre parallelism	Drafting roller drives; relatively minor energy contributor
Roving	Further drafts the sliver and inserts a small amount of twist; wound on a bobbin for transport to the ring frame	Flyer and bobbin drive; minor energy contributor
Ring spinning (ring frame)	Drafts the roving to final fineness and inserts twist via the traveller-on-ring mechanism	Ring spindle drives and ring rail travel; the primary SEC-determining stage; strongly dependent on yarn fineness
Winding (autoconer)	Transfers yarn from small ring bobbins to larger conical packages; clears yarn faults	Winding drum and splicing motors; SEC increases for finer yarns due to higher number of splices per kg

1.3 System boundary

The system boundary covers gate-to-gate ring spinning: from raw fibre input to finished yarn at the spinning mill gate. The following flows are **included**:

- Direct and auxiliary electricity (all sub-processes)
- Lubricating oil input and waste mineral oil output
- Process water (MMCF and wool routes only)
- Plastic bobbin input and waste polypropylene output
- Capital goods: machine amortisation and EoL steel
- Building infrastructure

The following are explicitly **excluded**:

- Fibre production (upstream)
- **Yarn losses and fibre waste**: handled externally via the IOR. No yarn or fibre input row and no textile waste row appear in any dataset
- Downstream processing (weaving, knitting, finishing)
- Wastewater discharge (currently absent; acknowledged as a limitation)

1.4 Functional unit and scope

Parameter	Value
Functional unit	1 kg of finished ring-spun yarn at the spinning mill gate
Background database	ecoinvent 3.12, system model: cut-off by classification
Background linking	Generic market activities, geography GLO or RoW unless process-specific geography is required
Impact method	IPCC 2021 GWP100 via Brightway2
Grid dimensions	15 fibre families × up to 46 dtex points = 875 individual process datasets
dtex range	45–600 dtex in industrially relevant steps
Foreground flows	Electricity, lubricant (input + waste mineral oil), process water (MMCF and wool), plastic bobbin (input + waste), capital goods (machine + building)
Yarn loss convention	Handled externally via IOR; no inline yarn/fibre input or textile waste row

2. Data sources and literature review

2.1 Review of existing LCI databases

Existing spinning datasets in EF 3.1 and ecoinvent 3.12 provide electricity values for a limited set of material types and yarn counts, but without transparent documentation of the underlying

regression or measurement basis. The EF 3.1 spinning datasets cover cotton and some synthetic yarns but do not parameterise electricity as a continuous function of yarn fineness, and they do not distinguish between carded and combed preparation routes. The Ecobalyse parametric model (ADEME, 2024) provides a Nm-based linear formula but is not validated against independent measured data within its documentation.

This absence of transparent, fineness-resolved, multi-material spinning LCIs is the primary motivation for constructing the Carbonfact ring-spinning model from primary literature sources.

2.2 Primary literature sources for ring-spinning electricity

2.2.1 Ecobalyse parametric model (ADEME & Ministère de la Transition Écologique, 2024)

Ecobalyse models ring-spinning electricity as a linear function of yarn title in metric count (Nm):

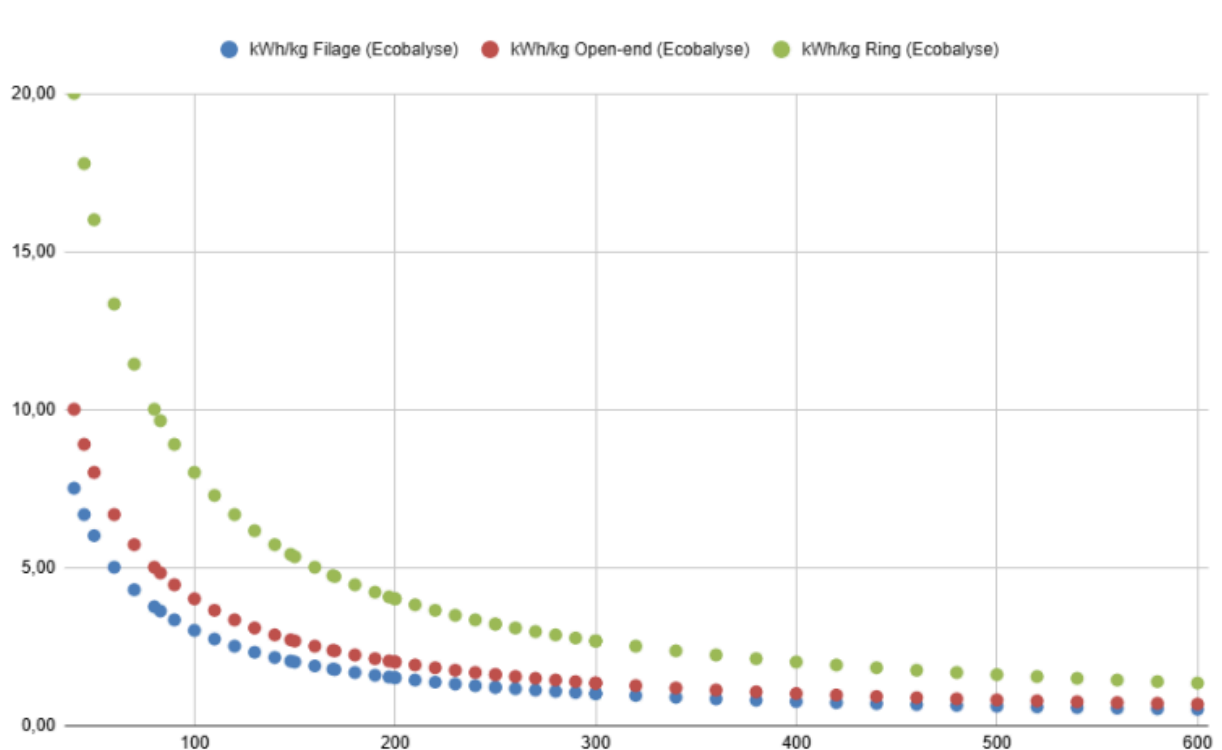
$$\text{kWh}(\text{ring spinning}) = \frac{\text{Titration [Nm]}}{50} \times C_{\text{ring}} \times \text{Outgoing mass [kg]}$$

$$C_{\text{ring}} = 4 \text{ kWh/kg at } 50 \text{ Nm (reference count, } \approx 200 \text{ dtex)}$$

Simplifies to:

$$\text{SEC}_{\text{Ecobalyse}}(\text{dtex}) = \frac{800}{\text{dtex}} \quad [\text{kWh/kg}]$$

It serves as one of three sources blended into the Carbonfact cotton ring curve (Section 3.1.5).



2.2.2 Koç & Kaplan (2007)

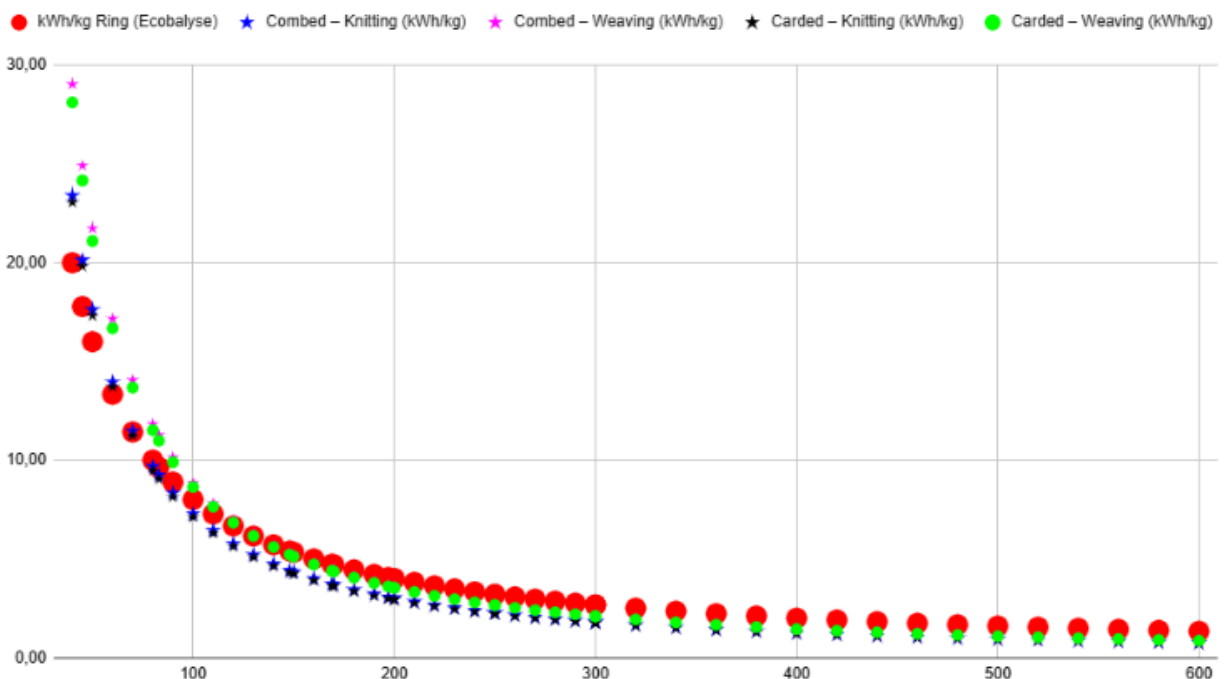
Koç & Kaplan (2007) report measured specific electricity consumption (kWh/kg) for ring-spun cotton yarn across a range of yarn titles, distinguishing between carded and combed preparation routes and between knitting and weaving end-uses. This four-category dataset provides the most granular empirically measured breakdown for ring spinning in the reviewed literature. Key values at 20 tex (≈ 200 dtex) are:

Category	SEC at 20 tex (≈ 200 dtex) [kWh/kg]
Combed – knitting	3.06
Combed – weaving	3.64
Carded – knitting	2.96
Carded – weaving	3.57

Power-law functions of the form $SEC(dtex) = A \times (dtex/10)^b$ were fitted to the published data by log-log regression, yielding:

Category	A	b	R ² (over published range)	Empirical domain (dtex)
Combed – knitting	136.84	-1.274	≈ 0.99	120–370
Combed – weaving	175.31	-1.297	≈ 0.99	120–370
Carded – knitting	136.50	-1.283	≈ 0.99	120–370
Carded – weaving	167.68	-1.288	≈ 0.99	120–370

Ecobalyse vs Koç 2007 extrapolated (45 to 600 dtex)



The exponents cluster tightly around **-1.28**, confirming the sharply increasing SEC at finer counts. The empirical domain of these fits is 120–370 dtex; outside this range the equations are used as controlled extrapolations.

2.2.3 Laursen et al. (2007) — EDIPTEX

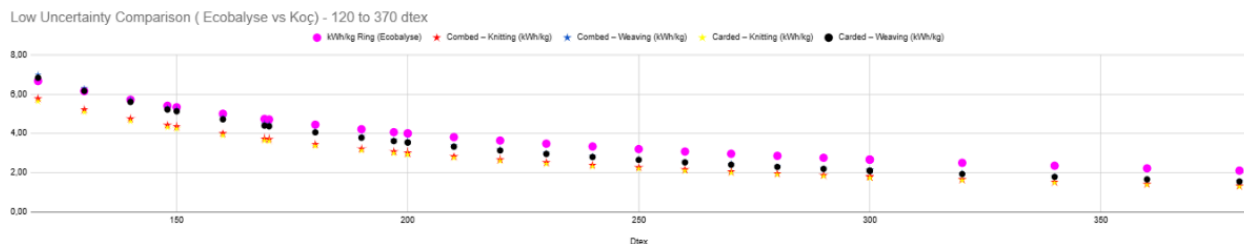
Laursen et al. (2007) (EDIPTEX, Table 9.1) report energy consumption for ring-spun yarns across three material configurations — 100% combed cotton, 100% synthetic, and 65/35 PES/cotton — as a function of yarn title over 13–60 tex (130–600 dtex). The Laursen data include auxiliary energy from humidification and lighting, making them systematically higher than machine-only measurements. Converting to kWh/kg and dtex yields:

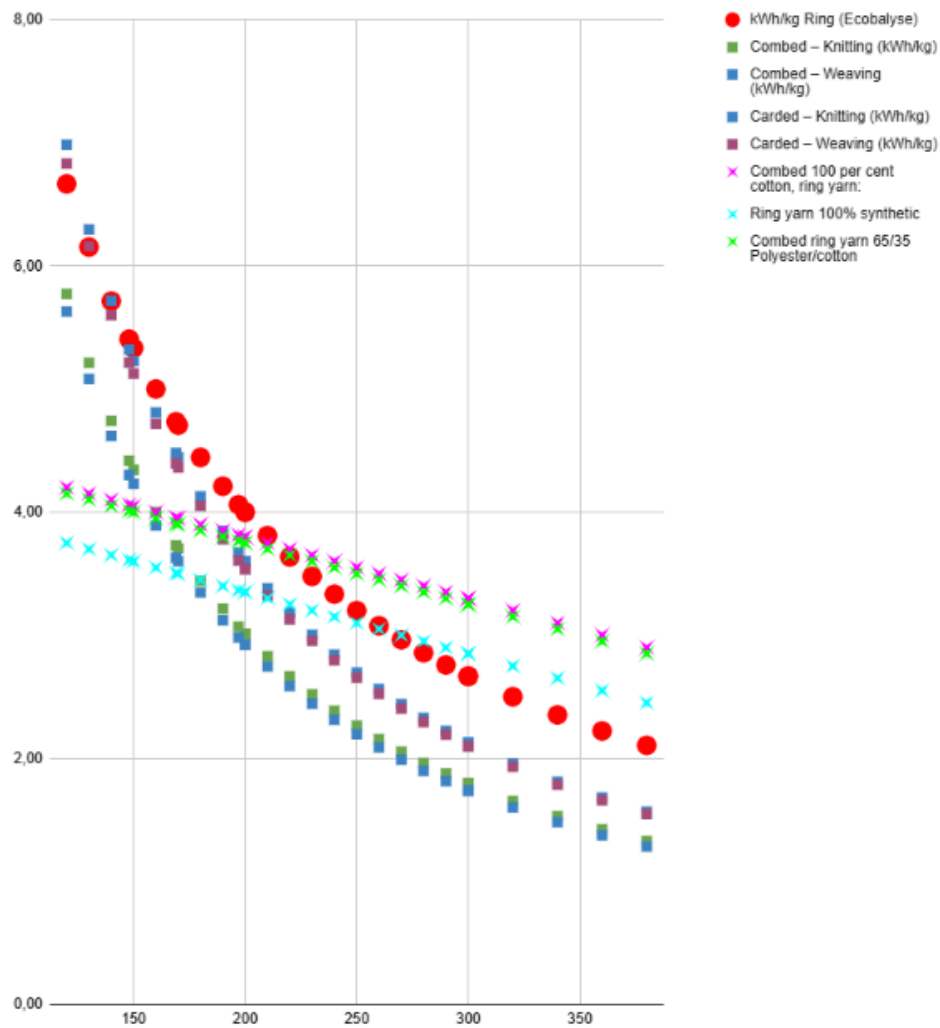
100% combed cotton: $SEC \approx 4.80 - 0.005 \times dtex$ [kWh/kg, valid ≈ 130 –600 dtex]

100% synthetic: $SEC \approx 4.35 - 0.005 \times dtex$

65/35 PES/cotton: $SEC \approx 4.75 - 0.005 \times dtex$

These linear curves are shallower than the Koç & Kaplan power-laws and represent a more comprehensive system boundary, making them a complementary rather than competing source.





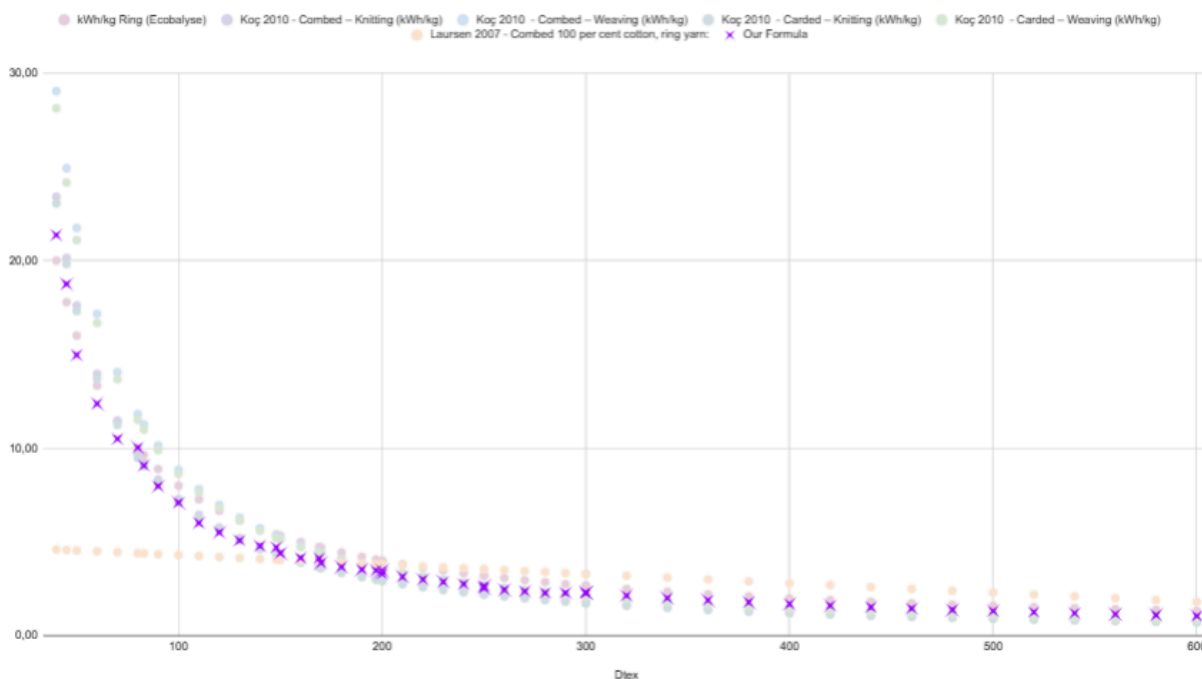
2.3 External validation sources

The blended Carbonfact cotton ring curve was cross-validated against the following independent anchor points, which were not used in the fitting procedure:

Source	dtex	SEC (kWh/kg)	Notes
Haque & Naebe (2024), Future Textiles, Deakin University	45	22.4	Independent measurement at the fine end; confirms high SEC for very fine counts
Alam et al. (2023), Tekstil ve Konfeksiyon	180	1.35	Substantially below the blended model (~3.9 kWh/kg); attributed to a high-efficiency mill or narrower system boundary (likely no humidification/auxiliary loads)
Laursen et al. (2007) — cotton formula at	170	4.51	Used as validation rather than fitting point

Source	dtex	SEC (kWh/kg)	Notes
170 dtex			
IDEMAT database (TU Delft, 2012)	250	4.72	Consistent with Laursen curve; slightly higher than blended model (~3.09 kWh/kg), suggesting IDEMAT includes auxiliary loads
Laursen et al. (2007) — cotton formula at 250 dtex	250	3.14	Lower bound at 250 dtex; blended model sits between the two 250-dtex anchors
Haque & Naebe (2024)	300	3.40	Within factor 1.4 of blended model (~2.56 kWh/kg)
Laursen et al. (2007) — cotton at 360 dtex	360	2.10	Excellent match with blended model (~2.12 kWh/kg)
Sandin et al. (2019), Mistra Future Fashion	600	3.80	Substantially above blended model (~1.05 kWh/kg); most likely includes upstream cotton-growing burdens or a wider system boundary

Ring Spun Formula by Carbonfact



2.4 Sources for non-electricity parameters

- **Lubrication:** Faysal et al. (2021) for cotton ring spinning oil ranges; LCA ReviWool (Manteco) for wool lanolin-based spinning oils; All/Made2Flow internal benchmark for fibre-family modifiers.

- **Yarn losses:** EDIPTEX (Laursen et al., 2007); Textile Exchange CFMB (2019); LCA ReviWool; mill-level case studies.
 - **Process water (MMCF and wool):** EDIPTEX Table 9.1; LCA ReviWool (Manteco).
 - **Capital goods:** Generic spinning line parameters; LCA ReviWool for worsted wool dedicated preparation line.
 - **Plastic bobbins:** BREF (European Commission, 2003); textile LCA case studies.
-

3. Life cycle inventory modelling

3.1 Electricity model

3.1.1 Conceptual basis

In ring spinning, specific electricity consumption (SEC, kWh/kg) increases as yarn becomes finer. The physical mechanism is straightforward: finer yarns require more drafting stages, more twist insertions per unit mass, and more machine time per kilogram at both the ring frame and winding stages. This inverse relationship between SEC and yarn fineness is well documented in the spinning literature and forms the guiding principle of the model.

3.1.2 Step 1 — Ecobalyse baseline for ring spinning

As an initial reference, the Ecobalyse parametric model was used to generate an SEC curve over 45–600 dtex:

$$SEC_{\text{Ecobalyse}}(\text{dtex}) = \frac{800}{\text{dtex}} \quad [\text{kWh/kg}]$$

Over the full range 45–600 dtex, this curve has an average SEC of approximately **5.29 kWh/kg**, decreasing monotonically from fine to coarse yarns.

3.1.3 Step 2 — Power-law fit to Koç & Kaplan (2007) data

Power-law functions were fitted to the Koç & Kaplan data by log–log regression:

$$SEC_{\text{Koç}}(\text{dtex}) = A \times \left(\frac{\text{dtex}}{10}\right)^b$$

The four fitted curves (combed knitting, combed weaving, carded knitting, carded weaving) are averaged to form a single Koç-based cotton curve used in the blending step.

3.1.4 Step 3 — Laursen et al. (2007) cotton ring linear curve

The EDIPTEX cotton ring linear expression:

$$SEC_{\text{Laursen,cotton}}(\text{dtex}) \approx 4.80 - 0.005 \times \text{dtex} \quad [\text{kWh/kg, valid } \approx 130\text{--}600 \text{ dtex}]$$

This curve is used as the third component in the blending step (120–600 dtex zone only).

3.1.5 Step 4 — Constructing the blended Carbonfact cotton ring curve

To avoid relying on a single source, a blended SEC curve was constructed for 100% cotton ring-spun yarn using a two-zone approach:

dtex range	Sources included	Rationale
45–120 dtex (fine yarns)	Ecobalyse + Koç average (2 sources)	Laursen et al. valid range starts at 130 dtex; only two sources available
120–600 dtex (main industrial window)	Ecobalyse + Koç average + Laursen cotton (3 sources, equal weights)	All three sources within their most robust domain; three-way averaging reduces dependence on any single model artefact

Indicative blended values:

dtex	SEC _{CF,cotton} blended (kWh/kg)	SEC_Ecobalyse	SEC_Koç avg	SEC_Laursen
45	≈ 21.4	17.78	≈ 25.1	— (outside range)
120	≈ 6.74	6.67	≈ 5.77	4.20
200	≈ 3.50	4.00	3.31	3.80
300	≈ 2.29	2.67	≈ 1.83	3.30
400	≈ 1.69	2.00	≈ 1.25	2.80
600	≈ 1.05	1.33	≈ 0.74	1.80

3.1.6 Step 5 — Fitted single power-law for the blended curve

For implementation in parametric tools, the discrete blended cotton curve was approximated by a single power-law fitted by log-log regression over all 46 dtex points (45–600 dtex):

$$\text{SEC}_{\text{CF,cotton}}(\text{dtex}) = 977.3 \times \text{dtex}^{-1.132} \quad [\text{kWh/kg}]$$

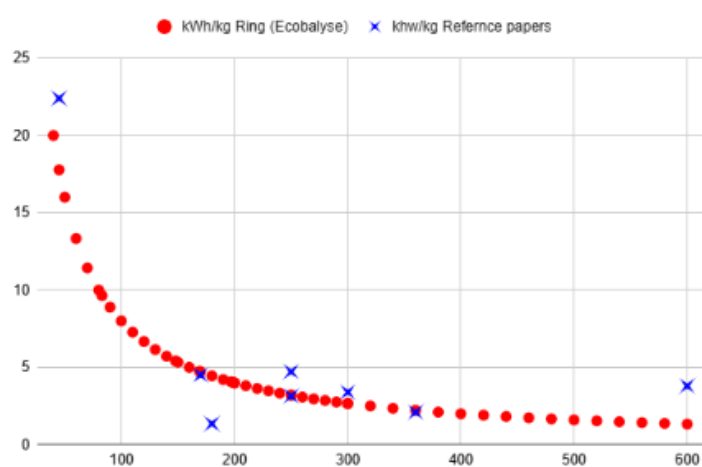
$R^2 = 0.998$ against the blended lookup table

Key reproduced values confirming fit quality:

- dtex = 45: $\text{SEC} = 977.3 \times 45^{-1.132} \approx 19.5 \text{ kWh/kg}$ (blended table: ~21.4 kWh/kg; fine-yarn extrapolation zone)
- dtex = 200: $\text{SEC} = 977.3 \times 200^{-1.132} \approx 3.48 \text{ kWh/kg}$ (blended table: ~3.50 kWh/kg)
- dtex = 600: $\text{SEC} = 977.3 \times 600^{-1.132} \approx 1.04 \text{ kWh/kg}$ (blended table: ~1.05 kWh/kg)

Clamping rules:

- dtex < 45: apply SEC at 45 dtex; document as extrapolation with ≥50% uncertainty
- dtex > 600: apply SEC at 600 dtex unless primary mill data are available



3.1.7 Extension to other fibre families via material-type factors

For non-cotton fibre families, SEC is derived by multiplying the cotton baseline by a dimensionless material factor f_{material} :

$$\text{SEC}_{\text{material}}(\text{dtex}) = f_{\text{material}} \times \text{SEC}_{\text{CF, cotton}}(\text{dtex})$$

Material type	f_{material}	Basis	Approx. SEC at 200 dtex (kWh/kg)
Cotton (all ring routes)	1.00	Reference material	3.48
Acrylic, Modacrylic, PAN	1.57	All/Made2Flow; elevated friction and static	5.46
Cotton / Man-Made (50/50 blends)	1.36	All/Made2Flow	4.73
CVC (cotton-rich ~50/50 cotton-PES)	1.21	All/Made2Flow	4.21
Elastane (Spandex/Lycra, TPU)	2.14	All/Made2Flow; highest factor; requires very fine counts and high machine speeds	7.44
Linen blends	1.60	All/Made2Flow; linen fibre stiffness increases spinning difficulty	5.57
MMCF (Rayon, Viscose, Lyocell)	1.29	All/Made2Flow	4.49 + process water
PC (Polyester/Cotton 50/50)	1.29	All/Made2Flow	4.49
Poly blends (generic synthetics)	1.36	All/Made2Flow	4.73

Zone	dtex range	Recommended uncertainty band	Primary sources of uncertainty
			system boundary ambiguity
Non-cotton material factors	All dtex	Additional ±20–30% on top of cotton uncertainty	Single unpublished All/Made2Flow source; no independent verification

3.2 Lubrication model

Lubricant consumption in ring spinning depends on fibre type, yarn fineness, and machine configuration. Three distinct profiles are applied, each grounded in specific literature sources.

3.2.1 Cotton baseline — decreasing function of dtex

For cotton ring yarn and most blends, lubricant consumption follows a linear decreasing function of dtex:

$$L_{\text{cotton}}(\text{dtex}) = 0.00200 - \frac{(0.00200 - 0.00050)}{(600 - 45)} \times (\text{dtex} - 45) \quad [\text{kg/kg}]$$

Fibre family	Literature range (kg/kg)	Observation	References
Cotton	0.0018–0.0035 ml/kg ≈ 0.0002–0.00035 kg/kg	Range 0.18–0.35 ml/kg from ring-spinning tribology studies; production value uses mid-range decreasing with dtex	Faysal et al. (2021)
Acrylic, Modacrylic, PAN	0.0025–0.005 kg/kg	Higher frictional demands of acrylic fibre in ring spinning; scaled upward relative to cotton baseline; decreasing with dtex	Fibres, A. A. (2004); Kent (2013)
CVC, PC, Cotton/Man-Made	±10–20% of cotton baseline	Cotton-rich blends follow the same decreasing profile with small modifiers reflecting blend ratio	Faysal et al. (2021)

3.2.2 Fixed low value — elastane, MMCF, polyamides, silk

Elastane (Spandex/Lycra), MMCF (Rayon, Viscose, Lyocell), Polyamides, and Silk blends all use a **fixed lubricant value of 0.000144 kg/kg**, constant across all dtex values. This value is sourced from the JRC Supporting Document for the LCA-LCC model (European Commission, 2024), which reports 0.144 g/kg of yarn as the standard lubricant floor value for synthetic/cellulosic ring-spun routes. Although the SEC factor for elastane is the highest in the matrix (\$f = 2.14\$), its lubricant demand is decoupled from SEC because elastane spinning uses a dry-spinning or wet-spinning route with different tribological requirements from cotton ring frames.

⚠ Limitation: This 0.000144 kg/kg value (JRC, 2024) has not been independently verified against primary mill data for ring spinning of these fibre types. It should be treated as a lower-bound placeholder and updated when mill data become available.

3.2.3 Fixed high value — polyester

Polyester uses a **fixed lubricant value of 0.010 kg/kg**, constant across all dtex values. This is substantially higher than cotton at comparable counts, reflecting the heavy spin finish required to control static and friction on synthetic fibre surfaces. The value is consistent with AFIRM Group guidance on spin finish application rates for polyester ring spinning (Nautiyal et al., 2025; AFIRM Group, 2022). **This value should be verified against polyester ring-spinning mill data.**

3.2.4 Fixed high value — wool (both routes)

Both Worsted Wool and Woollen Wool use a **fixed lubricant value of 0.0566 kg/kg**, constant across all dtex values — approximately 28–100 times higher than the cotton baseline. This reflects the application of wool spinning oils (lanolin-based or synthetic equivalents) applied during carding and combing. Sourced from LCA ReviWool (Manteco).

3.3 Process water model

Process water is zero for dry ring-spun routes (cotton, all synthetics, acrylic/PAN, poly blends, generic blends). Three fibre groups carry an explicit, non-zero process water input linked to `market for tap water {RoW} | tap water | cut-off`, decreasing with dtex:

Fibre family	Water at 50 dtex (kg/kg)	Water at 200 dtex (kg/kg)	Water at 400 dtex (kg/kg)	Source
MMCF (Rayon, Viscose, Lyocell)	~2.18	~1.59	~0.76	EDIPTEx (Laursen et al., 2007); wet preparation of cellulosic fibres
Woollen Wool	~2.11	~1.03	~0.61	LCA ReviWool (Manteco); scouring and wet preparation
Worsted Wool	~2.11	~1.03	~0.61	LCA ReviWool (Manteco); as woollen route

Polyamides and Polyester carry a trace value of 8.33×10^{-5} kg/kg (model placeholder; not measured). No wastewater discharge flow is currently modelled — acknowledged as a limitation in Section 5.

3.4 Sizing

Sizing (starch or synthetic binder application to warp yarns for weaving) is **not included** in the ring-spinning LCI. Sizing is a downstream process that applies specifically to weaving

preparation and is not a function of ring spinning itself. It may be included in a dedicated weaving LCI as an upstream step if required by the allocation approach.

3.5 Yarn loss and Input–Output Ratio (IOR)

Yarn waste and losses per kilogram of finished yarn are not modelled inline. They are communicated to the caller via the Input–Output Ratio (IOR), which scales the upstream fibre input accordingly.

$$\text{IOR} = \frac{1}{1 - L}$$

where L is the fibre/yarn loss rate expressed as a fraction $(0 < L < 1)$.

For every 1 kg of finished yarn, the caller must supply $\text{IOR} \times 1$ kg of upstream fibre. The lost fraction $(\text{IOR} - 1)$ kg per kg of finished yarn is routed by the caller to `market for waste yarn and waste textile {RoW} | cut-off`.

Representative IOR values at 200 dtex:

Material / Route	Literature range (loss %)	Loss at 200 dtex (%)	IOR (kg/kg)	Observation	References
Average / Cotton baseline	18–22%	18.62%	1.2288	Opening/carding losses + spinning waste; average across cotton routes	Laursen et al. (2007); JRC (2024): 22%; Wendin (2007): 43% upper bound
Cotton Combed – Knitting	25–35%	17.21%	1.2079	Combing noil (12–15% of input) + carding and spinning losses; higher waste than carded due to noil	EDIPTEx Table 16.1 (EDIPTEx, 2007); SITRA (2021)
Cotton Combed – Weaving	25–35%	17.21%	1.2079	As combed knitting	EDIPTEx (2007); SITRA (2021)
Cotton Carded – Knitting	10–20%	32.21%	1.4751	No combing noil; high opening/carding losses; high breakage at fine counts	EDIPTEx (2007); SITRA (2021)

Material / Route	Literature range (loss %)	Loss at 200 dtex (%)	IOR (kg/kg)	Observation	References
Cotton Carded – Weaving	10–20%	32.21%	1.4751	As carded knitting	EDIPTX (2007); SITRA (2021)
Acrylic / Modacrylic / PAN	8–13%	10.90%	1.1223	Lower preparation losses than cotton; spinning waste dominates; good compatibility with ring frame	Altun (2012); Sen et al. (1996)
MMCF (Rayon, Viscose, Lyocell)	3–7%	5.90%	1.0627	No opening/carding for manufactured fibres; minimal preparation losses; same assumption as polyester per literature	JRC (2024); Babel et al. (2019)
Polyester (PET, PBT, PTT, PLA)	3–7%	5.90%	1.0627	Very low losses; OE runs well with uniform synthetic fibres; ring values consistent	Szonei (2024); Textile Exchange CFMB (2019)
CVC (cotton-rich ~50/50)	9–13.5%	~12%	~1.14	Cotton 15–20% / PES 3–7% → 50/50 blend gives 9–13.5%; scalable to other blend ratios	MacDonald & Whitley (2009); Textile School; Indian Textile Journal
PC (polyester-cotton 50/50)	9–13.5%	~12%	~1.14	As CVC; can be calculated for any composition (35/65, 40/60, etc.)	MacDonald & Whitley (2009)
Cotton/Man-Made	9–13.5%	~12%	~1.14	Same approach as CVC; weighted by composition; MMCF	Internal calculation; Textile Exchange CFMB

Material / Route	Literature range (loss %)	Loss at 200 dtex (%)	IOR (kg/kg)	Observation	References
				equivalent to PES for loss purposes	
Elastane (Spandex/Lycra)	1–3%	1.49%	1.0151	Very low loss; minimal mechanical processing; almost exclusively filament route	Internal calculation
Spun Silk	40–60%	54.48%	2.1968	Reeling, degumming, combing losses dominate; very high and variable	Sashina & Yakovleva (2023)
Worsted Wool	5–30%	8.00%	1.0870	Wet preparation, carding, and combing losses; LCA ReviWool reference for ring worsted	Textile Exchange CFMB (2019)
Woollen Wool	5–30%	18.00%	1.2195	Wet preparation and carding losses; no combing; LCA ReviWool reference	Textile Exchange CFMB (2019)

The full IOR table covering all 875 dataset combinations is maintained in [Spinning_IORvalues.xlsx](#).

3.6 Capital goods: machinery and buildings

3.6.1 Spinning machinery — derivation

$Q_{\text{lifetime}} = \text{service life [yr]} \times \text{operating days [days/yr]} \times \text{operating hours [h/day]} \times \text{throughput [kg/h]}$

Default (average scenario): $Q_{\text{lifetime}} = 15 \times 300 \times 16 \times \text{throughput} = 2,155,172 \text{ kg}$

Capital goods intensity (standard variant): $\frac{1}{Q_{\text{lifetime}}} = \frac{1}{2,155,172} \approx$

$$4.64 \times 10^{-7} \text{ unit/kg yarn}$$

EoL steel: $\frac{2,800 \text{ kg}}{2,155,172 \text{ kg}} \approx 0.0013 \text{ kg/kg yarn}$

Three variant groups are defined, reflecting differences in machine type and preparation infrastructure:

Variant group	Capital goods (unit/kg)	EoL steel (kg/kg)	Rationale
Standard (cotton, synthetics, blends, MMCF, elastane, silk)	4.64×10^{-7}	0.0013	Generic ring spinning + winding line, 2,800 kg
Cotton Combed and Carded routes + Woollen Wool	1.25×10^{-6}	0.0035	Combined combing preparation + ring spinning line; higher total machine mass per kg output
Worsted Wool	2.21×10^{-6}	0.00620	Dedicated worsted preparation (scouring, carding, combing, drawing) + ring spinning; sourced from LCA ReviWool

Shared methodology: Capital Goods & End-of-Life (Machinery) in LCA — Full derivation, lifetime scenarios, EoL split. This section states only ring-spinning-specific values.

3.6.2 Building infrastructure

Shared methodology: Building Infrastructure in LCA — Process-specific value: $1.0 \times 10^{-5} \text{ m}^2/\text{kg}$, constant across all fibre families and dtex. Activity: `building, hall {GL0} | building, hall | cut-off`.

3.7 Plastic bobbins

Plastic carriers (bobbins, cheeses) are modelled with a constant input of **0.001 kg of polypropylene per kilogram of yarn**, linked to `market for polypropylene, granulate {GL0} | polypropylene, granulate | cut-off`. The same mass is output as waste polypropylene assumed to enter a recycling process: `waste polypropylene, for recycling, unsorted {GL0} | Recycled Content cut-off`. Constant across all fibre families and dtex values, consistent with BREF (European Commission, 2003).

3.8 Dataset naming convention

All 875 datasets follow the naming convention: `Spinning to yarn, [Fibre family], [dtex] dtex, Ring Spinning`. Example: `Spinning to yarn, Cotton, 200 dtex, Ring Spinning`.

4. Life cycle impact assessment

4.1 LCIA results

Climate change impacts (kg CO₂eq / kg yarn) were computed using Brightway2 with the IPCC 2021 GWP100 characterisation factors, applied to the foreground inventory with ecoinvent 3.12 (cut-off) background processes and the GLO average electricity mix. The full dataset spans 90 process variants across 10 fibre families and 9 dtex points (45–370 dtex); a representative selection is shown below. Full results across all 16 EF 3.1 impact indicators are available in [impact-scores.csv](#).

Cotton routes — combed knitting, full dtex range

Process	GHG (kg CO ₂ eq / kg yarn)
Cotton Combed – Knitting, 45 dtex	13.04
Cotton Combed – Knitting, 70 dtex	7.90
Cotton Combed – Knitting, 120 dtex	4.31
Cotton Combed – Knitting, 200 dtex	2.45
Cotton Combed – Knitting, 300 dtex	1.58
Cotton Combed – Knitting, 370 dtex	1.26
Cotton Combed – Weaving, 200 dtex	2.65
Cotton Combed – Weaving, 370 dtex	1.34

Synthetic and man-made fibres — 200 dtex reference count

Process	GHG (kg CO ₂ eq / kg yarn)
Acrylic, Modacrylic, PAN, 200 dtex	3.78
Elastane (Spandex/Lycra), 200 dtex	5.13
MMCF (Rayon, Viscose, Lyocell), 200 dtex	3.84
Polyamides (Nylon 6, 6.6, PA 11, PA 12, ...), 200 dtex	3.60
Polyester (PET, PBT, PTT, PLA etc.), 200 dtex	2.79

Wool and silk — 200 dtex reference count

Process	GHG (kg CO ₂ eq / kg yarn)
Worsted Wool, 200 dtex	3.01
Woollen Wool, 200 dtex	2.45
Spun Silk, 200 dtex	4.29

Results confirm the expected inverse relationship between yarn fineness and GHG intensity across all material families. Elastane dominates at fine counts (31.4 kg CO₂eq/kg at 45 dtex) while woollen wool and combed cotton knitting converge at coarser counts (~1.3–1.4 kg CO₂eq/kg at 370 dtex). The GLO average electricity mix is the primary driver in all cases; substituting a regional grid will shift absolute values proportionally while preserving relative rankings across fibre families.

4.2 Comparison with EF 3.1

The Carbonfact ring-spinning model differs from EF 3.1 spinning datasets in four key structural ways. First, the Carbonfact model provides a continuous SEC function of dtex for 15 fibre families, whereas EF 3.1 datasets cover a limited set of material types at fixed counts. Second, the Carbonfact model separates carded and combed cotton routes with distinct loss rates and capital goods intensities; EF 3.1 does not. Third, the Carbonfact model includes explicit lubrication profiles — including the high fixed values for wool (0.0566 kg/kg) and polyester (0.010 kg/kg) — that are absent from EF 3.1 spinning entries. Fourth, the Carbonfact model documents all IOR values explicitly per material family and dtex, whereas EF 3.1 aggregates fibre losses into a single upstream factor.

For cotton at 200 dtex, the Carbonfact model yields approximately 2.45 kg CO₂eq/kg yarn (GLO electricity mix), compared with EF 3.1 values in the range of 2.21 kg CO₂eq/kg for similar configurations.

5. Inferences and limitations

Several limitations should be acknowledged when using this model.

The electricity model does not separate machine loads from auxiliary loads (compressors, HVAC, humidification, lighting) at the plant level. The blended SEC values include auxiliaries to the extent that they are present in the Laursen et al. source, but the exact split varies across mills. The model also does not distinguish between technology generations: modern compact spinning machines may consume 15–25% less electricity than classical ring frames at the same count.

The material-type SEC factors (Section 3.1.7) are derived from a single unpublished industry dataset (All/Made2Flow, 2023) and cannot be independently verified. They should be treated as central estimates with substantial uncertainty (± 20 –30%). The lubricant model for elastane, MMCF, polyamides, and silk uses a fixed floor value (0.000144 kg/kg) that may not reflect actual spinning-specific lubrication; similarly, the polyester fixed value (0.010 kg/kg) has not been verified against primary mill data.

Process water is modelled for MMCF and wool routes based on literature sources; no wastewater discharge flow is currently included, which means downstream treatment burdens are absent. The Alam et al. (2023) anchor at 180 dtex (1.35 kWh/kg) is substantially below the blended model (~3.9 kWh/kg), most likely due to a narrower system boundary or a high-efficiency mill not representative of global average conditions.

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7. Annex

LCI for 1 kg of ring-spun yarn — Cotton, 200 dtex (standard variant)

Flow	Value	Unit	Source / Rationale	
Electricity (input)	3.479	kWh	$SEC_{CF,cotton}(200) = 977.3 \times 200^{-1.132} = 3.479 \text{ kWh/kg}$. `market group for electricity, medium voltage {GLO} \	electricity, medium voltage \

Flow	Value	Unit	Source / Rationale	
Lubricating oil (input)	0.000303	kg	$L_{\text{cotton}}(200) = 0.00200 - 0.000271 \times (200 - 45)/100 = 0.000303$ kg/kg. `lubricating oil production {RoW} \	lubricating oil \
Waste mineral oil (output)	0.000303	kg	Equal to lubricant input. `market for waste mineral oil {RoW} \	waste mineral oil \
Polypropylene bobbin (input)	0.001	kg	Constant across all variants. `market for polypropylene, granulate {GLO} \	polypropylene, granulate \
Waste PP bobbin (output)	0.001	kg	Assumed sent to recycling. `waste polypropylene, for recycling, unsorted {GLO} \	Recycled Content cut-off`
Capital goods – machine (input)	4.64×10^{-7}	unit	$1/2,155,172 = 4.64 \times 10^{-7}$ unit/kg. `market for building machine {GLO} \	building machine \
EoL machine – steel (output)	0.0013	kg	$2,800/2,155,172 = 0.0013$ kg/kg. `market for waste steel {RoW} \	waste steel \
Building infrastructure (input)	1.0×10^{-5}	m ²	Constant. `building, hall {GLO} \	building, hall \
Process water	0	kg	Zero for dry cotton ring-spun route.	
Yarn / fibre input	— (IOR)	kg	Not modelled inline. Caller supplies $\text{IOR} \times 1$ kg upstream fibre. IOR at 200 dtex = 1.2288 (loss = 18.62%). Lost fraction (0.2288 kg/kg) routed to `market for waste yarn and waste textile {RoW} \	cut-off`.

LCI for 1 kg of ring-spun yarn — MMCF (Rayon, Viscose, Lyocell), 200 dtex

Flow	Value	Unit	Source / Rationale	
Electricity (input)	5.566	kWh	$\text{SEC}_{\text{MMCF}}(200) = 1.29 \times 3.479 = 5.566$ kWh/kg. `market group for electricity, medium voltage {GLO} \	cut-off`
Lubricating oil (input)	0.000144	kg	Fixed value, constant across all dtex. `lubricating oil production	cut-off`

Flow	Value	Unit	Source / Rationale	
			{RoW} \	
Waste mineral oil (output)	0.000144	kg	Equal to lubricant input. `market for waste mineral oil {RoW} \	cut-off`
Polypropylene bobbin (input)	0.001	kg	As cotton. `market for polypropylene, granulate {GLO} \	cut-off`
Waste PP bobbin (output)	0.001	kg	As cotton. `waste polypropylene, for recycling, unsorted {GLO} \	Recycled Content cut-off`
Capital goods – machine (input)	4.64×10^{-7}	unit	Standard variant. `market for building machine {GLO} \	cut-off`
EoL machine – steel (output)	0.0013	kg	`market for waste steel {RoW} \	cut-off`
Building infrastructure (input)	1.0×10^{-5}	m ²	`building, hall {GLO} \	cut-off`
Process water (input)	1.586	kg	Decreasing with dtex: ~2.18 kg/kg at 50 dtex → ~0.76 kg/kg at 400 dtex. Source: EDIPTEX (Laursen et al., 2007). `market for tap water {RoW} \	tap water \
Yarn / fibre input	— (IOR)	kg	IOR = 1.0627 at 200 dtex (MMCF loss = 5.90%). Lost fraction (0.0627 kg/kg) routed to `market for waste yarn and waste textile {RoW} \	cut-off`.

LCI for 1 kg of ring-spun yarn — Worsted Wool, 200 dtex

Flow	Value	Unit	Source / Rationale	
Electricity (input)	3.896	kWh	$SEC_{\text{worsted}}(200) = 1.12 \times 3.479 = 3.896$ kWh/kg. `market group for electricity, medium voltage {GLO} \	cut-off`
Lubricating oil (input)	0.0566	kg	Fixed value, constant across all dtex. Source: LCA ReviWool (Manteco). `lubricating oil production {RoW} \	cut-off`
Waste mineral oil (output)	0.0566	kg	Equal to lubricant input. `market for waste mineral oil {RoW} \	cut-off`

Flow	Value	Unit	Source / Rationale	
Polypropylene bobbin (input)	0.001	kg	As cotton. `market for polypropylene, granulate {GLO} \	cut-off`
Waste PP bobbin (output)	0.001	kg	As cotton. `waste polypropylene, for recycling, unsorted {GLO} \	Recycled Content cut-off`
Capital goods – machine (input)	2.21×10^{-6}	unit	Dedicated worsted preparation line. Source: LCA ReviWool. `market for building machine {GLO} \	cut-off`
EoL machine – steel (output)	0.00620	kg	Heavier worsted machinery. `market for waste steel {RoW} \	cut-off`
Building infrastructure (input)	1.0×10^{-5}	m ²	`building, hall {GLO} \	cut-off`
Process water (input)	1.03	kg	Decreasing with dtex: ~2.11 kg/kg at 50 dtex → ~0.61 kg/kg at 400 dtex. Source: LCA ReviWool. `market for tap water {RoW} \	tap water \
Yarn / fibre input	— (IOR)	kg	IOR = 1.0870 at 200 dtex (worsted wool loss = 8.00%). Lost fraction (0.087 kg/kg) routed to `market for waste yarn and waste textile {RoW} \	cut-off`.

IOR table — all fibre families at 200 dtex

Material / Route	Loss (%)	IOR (kg/kg)	Literature range	References
Average / Cotton baseline	18.62%	1.2288	18–22%	Laursen et al. (2007); JRC (2024); Wendin (2007)
Cotton Combed – Knitting	17.21%	1.2079	25–35%	EDIPTEx (2007); SITRA (2021)
Cotton Combed – Weaving	17.21%	1.2079	25–35%	EDIPTEx (2007); SITRA (2021)
Cotton Carded – Knitting	32.21%	1.4751	10–20%	EDIPTEx (2007); SITRA (2021)
Cotton Carded – Weaving	32.21%	1.4751	10–20%	EDIPTEx (2007); SITRA (2021)
Acrylic / Modacrylic / PAN	10.90%	1.1223	8–13%	Altun (2012); Sen et al. (1996)

Material / Route	Loss (%)	IOR (kg/kg)	Literature range	References
MMCF (Rayon, Viscose, Lyocell)	5.90%	1.0627	3–7%	JRC (2024); Babel et al. (2019)
Polyester (PET, PBT, PTT, PLA)	5.90%	1.0627	3–7%	Szonei (2024); Textile Exchange CFMB (2019)
Polyamides (Nylon 6, 6.6, PA 11, PA 12)	5.90%	1.0627	3–7%	Textile Exchange CFMB (2019)
CVC (cotton-rich ≈50/50)	~12%	~1.14	9–13.5%	MacDonald & Whitley (2009)
PC (polyester–cotton 50/50)	~12%	~1.14	9–13.5%	MacDonald & Whitley (2009)
Elastane (Spandex/Lycra)	1.49%	1.0151	1–3%	Internal calculation
Spun Silk	54.48%	2.1968	40–60%	Sashina & Yakovleva (2023)
Worsted Wool	8.00%	1.0870	5–30%	Textile Exchange CFMB (2019); LCA ReviWool
Woollen Wool	18.00%	1.2195	5–30%	Textile Exchange CFMB (2019); LCA ReviWool